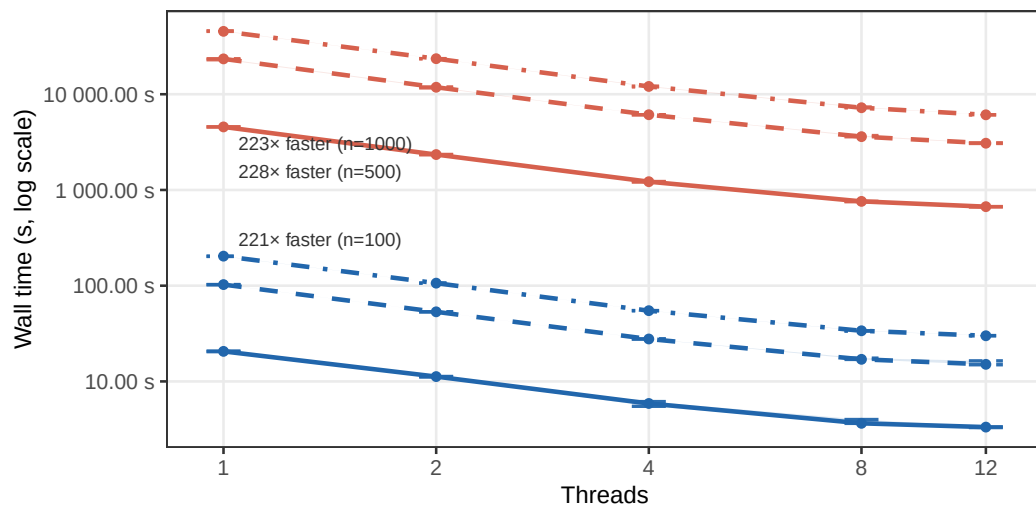


A

Runtime

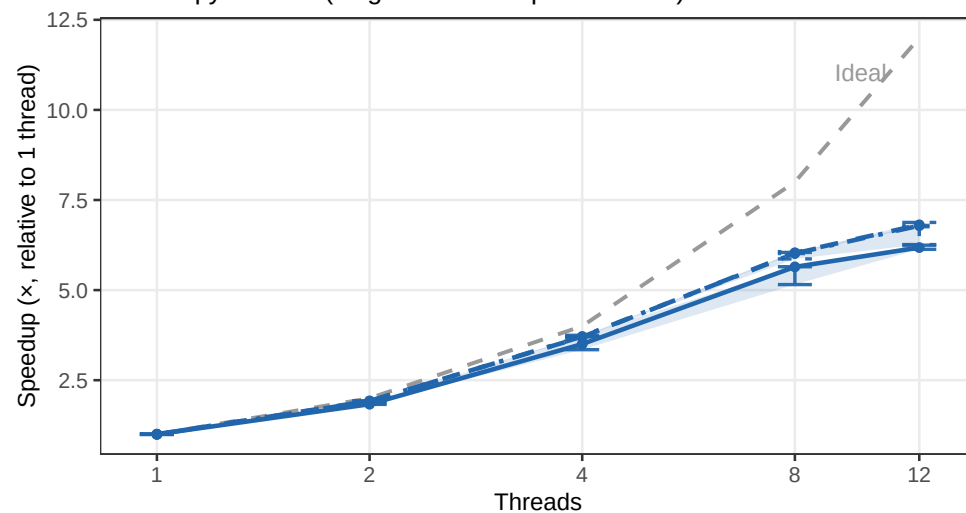
fast-mic vs COBRApy at matched core counts



B

fast-mic parallel speedup

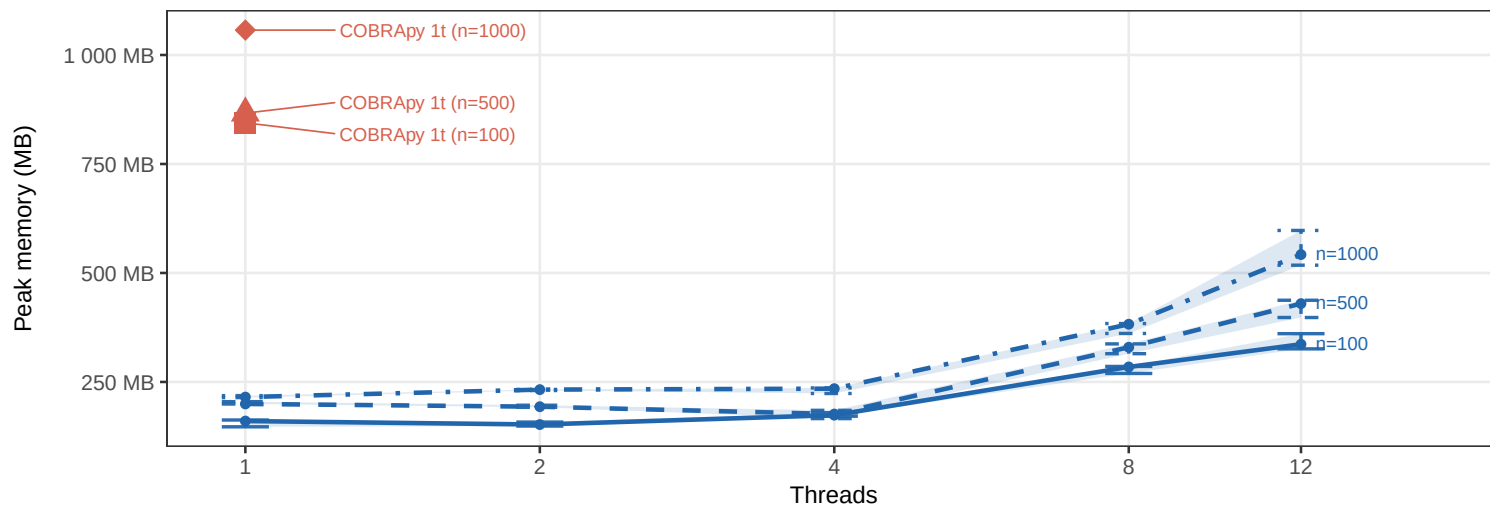
COBRApy omitted (single-threaded per FBA call)



C

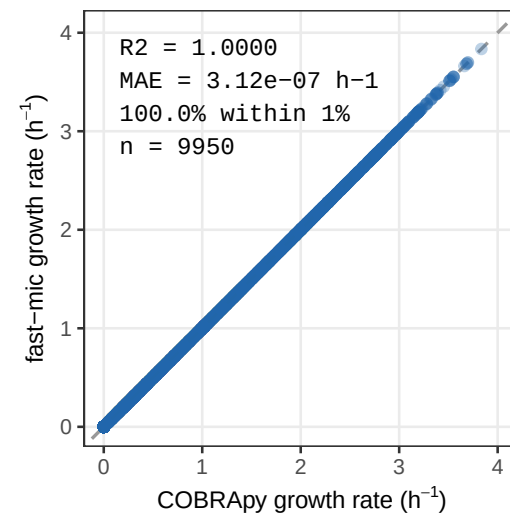
Memory usage

fast-mic curve; COBRApy shown only at 1 thread (true baseline)



D

Numerical agreement



Models — 100 — 500 - - 1000 Tool COBRApy fast-mic

Models — 100 — 500 - - 1000

Models ■ 100 ▲ 500 ◆ 1000